

Customer Churn Analysis – Telco Customer Churn

A Python-based exploratory data analysis (EDA) project using the **Telco Customer Churn** dataset. The project studies customer characteristics, subscription details, service usage, contract types, payment methods, and their relationship with customer churn.

Project scope: Exploratory Data Analysis and data preprocessing. The current notebook does **not** contain a machine-learning prediction model or model evaluation metrics.

Objective

The main objective is to understand customer churn patterns and identify customer attributes that are associated with higher churn so that the results can support future customer-retention analysis.

Dataset

- Records: **7,043 customers**
- Features: **21 columns**
- Target variable: Churn
- Main categories: demographics, tenure, phone/internet services, support services, contract, billing, payment method, and charges.

The notebook loads the dataset from Customer churn.csv. For GitHub, keep the dataset file in the project root if you have permission to redistribute it; otherwise, provide the original Kaggle source separately and place the downloaded CSV locally before running the notebook.

Technologies Used

- Python
- Pandas
- NumPy
- Matplotlib
- Seaborn
- Jupyter Notebook

Analysis Workflow

1. Import Python libraries.
2. Load the Telco customer churn CSV dataset.
3. Inspect the dataset structure and data types.
4. Check missing values and duplicate records.
5. Convert TotalCharges into numeric form after handling blank entries.
6. Convert SeniorCitizen from 0/1 into No/Yes labels.
7. Perform exploratory analysis using count plots, a pie chart, stacked percentage charts, and service-level visualizations.
8. Interpret churn patterns across customer characteristics and services.

Data Quality Checks

The notebook reports:

- **7,043 rows** and **21 columns**
- **0 null values** after preprocessing
- **0 duplicate rows**
- **0 duplicate customer IDs**

Key Findings

Overall churn

The dataset contains **1,869 churned customers (26.54%)** and **5,174 customers who did not churn (73.46%)**.

Gender

The analysis compares churn between male and female customers. Gender does not show a large difference in churn in the notebook's exploratory results.

Senior Citizen

The notebook shows a higher churn percentage for senior-citizen customers. The stacked percentage visualization shows approximately **41.7% churn among senior citizens** compared with **23.6% among non-senior customers**.

Tenure

Churn is more concentrated among customers with shorter tenure, indicating that the early customer relationship is an important period for retention analysis.

Contract type

The project compares **month-to-month, one-year, and two-year contracts**. Month-to-month customers show substantially more churn than customers on longer-term contracts.

Services

The notebook examines phone, internet, security, backup, device protection, technical support, and streaming services. The analysis highlights higher churn among **fiber-optic users** and among customers without some additional support/protection services.

Payment method

Four payment methods are compared: electronic check, mailed check, automatic bank transfer, and automatic credit card. The exploratory analysis shows comparatively high churn for **electronic-check users**.

Conclusion

The analysis indicates that customer churn is associated with multiple aspects of the customer relationship. Short tenure, month-to-month contracts, senior-citizen status, fiber-optic internet service, absence of selected support services, and electronic-check payment are among the patterns highlighted by the exploratory analysis.

These findings can be used as a starting point for a future churn-prediction model and customer-retention strategy analysis.

Current Limitations

The current notebook does not include:

- Train/test split
- Feature encoding for machine learning

- Classification algorithms
- Predicted churn labels
- Accuracy, precision, recall, or F1-score
- Confusion matrix
- ROC curve or AUC
- Feature-importance analysis

Future Scope

A future version can extend the project by applying classification algorithms such as Logistic Regression, Decision Tree, Random Forest, or other suitable models; comparing evaluation metrics; tuning hyperparameters; and identifying the most influential churn predictors.

Repository Structure

Customer-Churn-Analysis/

├── Customer_churn_analysis.ipynb

├── README.md

├── PROJECT_REPORT.md

└── images/

 ├── churn-percentage.png

 ├── churn-by-gender.png

 ├── churn-by-senior-citizen.png

 ├── churn-by-tenure.png

 ├── churn-by-contract.png

 ├── churn-by-services.png

 └── churn-by-payment-method.png

How to Run

1. Install Python 3.x and Jupyter Notebook/JupyterLab.
2. Install the required libraries:

`pip install pandas numpy matplotlib seaborn jupyter`

3. Put Customer churn.csv in the same directory as the notebook.
4. Open Customer_churn_analysis.ipynb in Jupyter Notebook or JupyterLab.
5. Run the cells from top to bottom.

Author

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